AI Development Manifestos and Frameworks

https://gemini.google.com/app/d81d36b4f586f7b3

User prompt: can you find similar manifestos especially in the AI developement, software factory world: https://github.com/twelve-factor/twelve-factor

Response: The original **Twelve-Factor App** methodology defined how cloud-native SaaS applications should be built. As software development shifts toward **AI-native engineering, autonomous agents, and AI software factories**, several open-source community manifestos and frameworks have emerged to adapt or expand those principles.

1. Direct "12-Factor" Adaptations for AI & Agents

- **12-Factor Agents (HumanLayer / GitHub)**
  - **The Premise:** A direct spiritual successor to the 12-factor app, written specifically for engineering teams building reliable LLM agents.
  - **Core Factors:**
    - *Factor 2: Own your prompts* — Keep prompts close to application code rather than black-box frameworks.
    - *Factor 3: Own your context window* — Explicitly manage what enters and leaves the context.
    - *Factor 5: Unify execution state and business state*.
    - *Factor 12: Make your agent a stateless reducer* — Treat prompt loops as deterministic state transitions where LLMs dictate next steps.
- **The 16-Factor App for AI (Google Cloud)**
  - **The Premise:** Keeps the original 12 cloud-native factors and adds four new principles for non-deterministic AI applications.
  - **The Four New Factors:**
    - *Factor XIII: Prompts as Code & Specs* — Version-control prompts, context-engineering logic, and golden evaluation datasets together.
    - *Factor XIV: State as a Service* — Externalize conversational memory into a dedicated store to keep application containers stateless.
    - *Factor XV: Observability for Non-Determinism* — Evaluate semantic output quality, token usage, and tool accuracy beyond basic HTTP status codes.
    - *Factor XVI: Evaluation-Driven CI/CD* — Automate regression testing against golden evaluation benchmarks before deploying prompt updates.

2. AI Software Factory & AI-Native Development Manifestos

- **Seven Rules for Building an AI-Native Software Factory (Pulumi)**
  - **The Premise:** Architectural guidelines for running software factories where AI agents write the code and engineers operate the factory.
  - **Key Rules:**
    - *Transform, don't enhance* — Rebuild development workflows from scratch rather than slapping AI co-pilots onto legacy steps.
    - *Pick tools your agents can drive* — Prefer type-safe APIs, CLIs, and declarative code over click-heavy web UIs.

    - *Design for convergence, not one-shot correctness* — Build iterative validation loops rather than expecting perfect generation on the first pass.

- **AI-Augmented Software Development Manifesto**
    - **The Premise:** Modeled after the Agile Manifesto, balancing rapid AI code generation with human judgment and architectural standards.
    - **Key Values:** Human agency and taste over agentic autopilot; tight human feedback loops over long unattended runs; treating code as cheap while system understanding is expensive.
- **The AI-Native Large-Scale Agile Software Development Manifesto (arXiv)**
    - **The Premise:** Rethinking large-scale agile methodologies for organizations running agent workforces.
    - **Core Principles:** *Parallel processes by default* (planning, building, and testing running concurrently), *Intent & Oversight teams* (engineers defining goals rather than typing lines of code), and *Living Knowledge* (version-controlled specs replacing decaying wiki pages).

3. Data & MLOps Foundations

Manifesto	Focus Area	Core Philosophy
Google's Rules of Machine Learning	ML Systems Engineering	43 practical rules emphasizing simple first pipelines, reliable metric design, logging features at serving time, and avoiding complex heuristics.
The DataOps Manifesto	Data Pipeline Factories	Adapts Lean manufacturing to data: treats analytics as code, demands disposable test environments, and mandates continuous quality monitoring.
The MLOps Manifesto	Model Lifecycle Operations	Focuses on end-to-end model reproducibility, governance accountability, continuous deployment, and statistical drift monitoring.

Google's Rules of Machine Learning Engineering Presentation

This presentation by Google research scientist Martin Zinkevich breaks down the essential software engineering principles for building reliable machine learning pipelines in production.

http://googleusercontent.com/youtube_content/1

User prompt: research more deeply and lets make an exhaustive list -

Response: Across the AI engineering landscape, the transition from **deterministic software** to **non-deterministic, agentic systems** has sparked a new wave of engineering manifestos, design standards, and operational frameworks.

Here is an exhaustive catalog of the primary manifestos, 12-factor extensions, and software factory principles governing modern AI development.

1. Direct "N-Factor" Methodologies for Agents & LLMs

These frameworks take the classic Twelve-Factor App methodology and adapt it to non-deterministic LLM loops, tool execution, and prompt engineering.

A. The 12-Factor Agent Methodology (12FA)

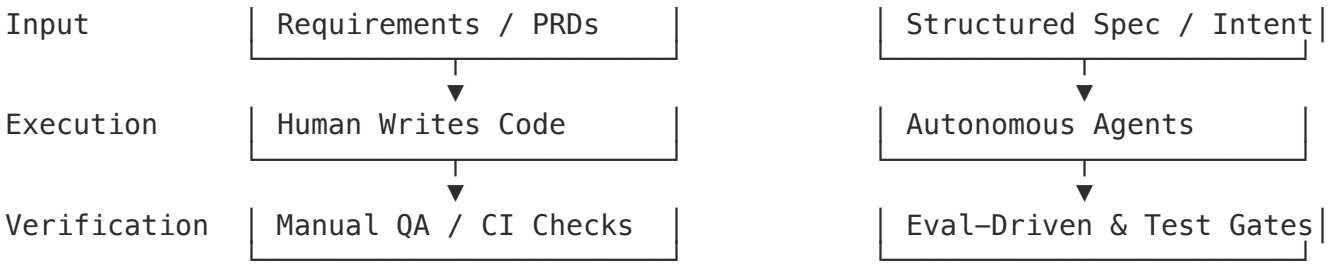
- **Origin:** HumanLayer / Community Open Source (Dexter Horthy)
- **Core Premise:** Moves agent development beyond fragile 70–80% prototype reliability into production-grade systems.
- **Key Principles:**
    1. **JSON Extraction as Foundation:** Treat LLMs primarily as structured data translators.
    2. **Own Your Prompts:** Keep prompts explicitly in application code; avoid heavy prompt abstractions or hidden magic.
    3. **Manage Context Windows Explicitly:** Actively prune, summarize, and manage context rather than blindly appending history.
    4. **Tools Are Just JSON & Code:** Treat tool calls as simple typed outputs validated via deterministic switch/case logic.
    5. **Own Your Control Flow:** Keep loops (OODA, ReAct) in native application code rather than delegating loop control to the model.
    6. **Stateless Agent Design:** Build execution steps to be idempotent and restartable to enable pause, resume, and horizontal scaling.
    7. **Separate Business State from Execution State:** Isolate transactional DB state from agent step history.
    8. **Contact Humans as First-Class Operations:** Treat human-in-the-loop routing as a standard tool call, not an exception or edge case.
    9. **Small, Focused Agents:** Keep agents bounded to 3–10 steps max; compose small agents rather than building monolithic chat agents.
    10. **Explicit Error Handling:** Compact execution errors into the next prompt to establish closed self-healing feedback loops.

B. The 16-Factor App for AI

- **Origin:** Google Cloud Architecture Team
- **Core Premise:** Retains all original 12 cloud-native factors and appends 4 factors tailored for probabilistic AI platforms.
- **The Additional Four Factors:**
    - **Factor XIII (Prompts as Code & Specs):** Version-control system prompts, context-engineering logic, and golden evaluation datasets in lockstep with application logic.
    - **Factor XIV (State as a Service):** Externalize conversational memory and vector state into specialized managed stores to ensure compute instances remain stateless.
    - **Factor XV (Observability for Non-Determinism):** Measure semantic drift, token efficiency, tool-selection accuracy, and hallucination rates alongside latency and HTTP errors.
    - **Factor XVI (Evaluation-Driven CI/CD):** Enforce automated evaluation regressions against benchmark datasets as a blocking deployment gate.

2. AI Software Factory & Spec-Driven Development (SDD)

These frameworks redefine how software is produced when agents perform the bulk of coding, testing, and infrastructure configuration.



A. Spec-Driven Development (SDD) Manifesto

- **Origin:** Microsoft / GitHub Spec Kit
- **Core Premise:** Inverts the legacy software paradigm: **Specification is the single source of truth; code is a secondary, generated artifact.**
- **The SDD Lifecycle Pipeline:**
    1. **Constitution:** Define immutable organizational guardrails, architectural standards, and security policies.
    2. **Specify:** Declare business intent, user scenarios, constraints, and explicit acceptance criteria.
    3. **Clarify:** Automatically resolve ambiguity and edge-case boundaries prior to code execution.
    4. **Plan & Task:** Translate intent into deterministic execution steps and architectural boundaries.
    5. **Implement & Validate:** Use AI agents to generate code and immediately evaluate outputs against the initial spec.

B. Seven Rules for an AI-Native Software Factory

- **Origin:** Pulumi / Gas City (Chris Sells)
- **Core Principles:**
    1. **Transform, Don't Enhance:** Rebuild workflows around agents from scratch rather than slapping AI plugins onto legacy human steps.
    2. **Remove the Problem, Don't Solve It:** Reframe architectural bottlenecks (e.g., spin up dedicated, isolated staging environments per agent run rather than building complex multi-tenant sandboxes).
    3. **Pick Tools Agents Can Drive:** Prefer type-safe APIs, CLIs, and strongly typed IaC (like TypeScript/Pulumi) over click-heavy web UIs or unstructured formats.
    4. **Don't Let One Agent Do Everything:** Delegate tasks across small, specialized agent roles with clear handoffs.
    5. **Measure Human Hours per Unit of Value:** Focus productivity metrics on reducing human intervention time rather than tracking raw token counts or lines of code.
    6. **Design for Convergence:** Build iterative feedback and self-correction loops rather than expecting one-shot code accuracy.
    7. **Run the Factory in the Cloud:** Keep agent execution, build systems, and evaluation environments cloud-hosted and ephemeral.

C. AI-Native Large-Scale Agile Software Development Manifesto

- **Origin:** Academic & Enterprise Engineering Consortium (arXiv:2605.07717)
- **Key Principles:**

- **Parallel Processes by Default:** Planning, codegen, testing, and security scanning run concurrently rather than in gated linear phases.
- **Intent & Oversight Teams:** Shift human software engineers from *code typists* to *intent architects and guardrail owners*.
- **Living Knowledge:** Replace static/decaying documentation with version-controlled, agent-readable contextual knowledge graphs.
- **Verification-First Assurance:** Enforce automated, continuous agent evaluation before merged changes reach production.

3. Security, Evaluation & MLOps Manifestos

Manifesto	Originator	Core Thesis / Philosophy
The Eval Manifesto	Braintrust	**"Evals are the new PRD."** AI products must be tested, not merely monitored. Traces are the atomic unit of AI observability, feeding directly into continuous evaluation datasets.
Shift to AI: Self-Protecting Software	Cycode	Replaces the static Software Development Lifecycle (SDLC) with the **Agentic Development Lifecycle (ADLC)**. Security must be embedded as active agentic defense operating in parallel with codegen.
Anthropic's Building Effective Agents	Anthropic	Prioritizes **simplicity, transparency, and Agent-Computer Interface (ACI) design**. Categorizes agent patterns into 6 execution topologies (Chain, Route, Parallel, Orchestrator-Worker, Evaluator-Optimizer, Autonomous).
Google Rules of Machine Learning	Google Engineering	Defines 43 rules for production ML pipelines: prioritize simple baseline heuristics, keep pipeline pipelines isolated, and treat feature engineering as code.
The DataOps Manifesto	DataOps.org	Applies Lean manufacturing to data pipelines: requires automated continuous testing, disposable dev environments, and treat data as code.

Summary Matrix: The Core Paradigm Shift

Dimension	Legacy Cloud-Native (Original 12-Factor)	AI Software Factory & Agent-Native Era
Source of Truth	Monolithic or Microservice Codebase	Versioned Specs, Prompts, & Context Archives
Primary Artifact	Executable Application Code	Executable Specifications & Eval Benchmarks
Execution Model	Deterministic Control Flow	Non-Deterministic Reasoning Loops & Tool Calling
Quality Control	Unit & Integration Testing	Semantic Evaluations (Evals) & Golden Datasets
Human Role	Writing logic & fixing syntax	Defining intent, reviewing diffs, & setting guardrails

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User prompt: now lets widen the net and find more principles for AI native software developement practices

Response: Widen the lens beyond 12-factor micro-rules, and you'll find that **AI-native software development is transforming the entire engineering lifecycle**. It reshapes how teams write code, manage context, design repositories, and verify results.

Below is an exhaustive matrix of the emerging paradigms, operational principles, and engineering manifestos defining this shift.

1. Operating Mindset & Human-Agent Roles

These principles define how human software engineers shift from *code typists* to *system orchestrators and taste directors*.

- **Humans *On* the Loop, Agents *In* the Loop**

  Taste, accountability, customer empathy, and responsibility are irreducibly human. No agent has skin in the game. Humans design the execution loops, set quality thresholds, and accept risk; agents execute within those boundaries.

- **The "First Mile" & "Last Mile" Rule**

  - **The First Mile (Design & Intent):** Framing the problem, setting constraints, and defining success.
  - **The Last Mile (Real-World Correctness):** Ensuring edge-case validity, security, and production fit. AI excels at the middle 80% (drafting logic, generating boilerplate, writing tests), but humans own the first and last miles.

- **Tokens Are Cheaper Than Human Hours**

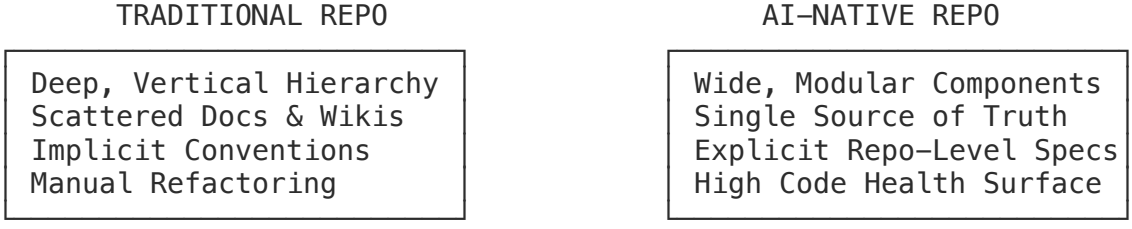
  Inference costs compound downward every quarter. Spend tokens liberally on evaluation runs, automated retries, spec verification, and self-healing loops. Cost anxiety should never be the reason a human performs manual labor that an agent can handle.

- **Software Is Clay**

  When code generation is nearly instant, software becomes highly malleable. Instead of over-architecting one "perfect" solution over months, ship ten quick iterations, evaluate real-world feedback, and reshape the system tomorrow.

2. Codebase Ergonomics & Context Engineering

Traditional codebases were designed to be readable by human eyes. AI-native codebases are designed to be **parsable, modular, and bounded for language models**.



- **Let AI "Dance in Chains" (Constraint Engineering)**

  Because LLMs are non-deterministic, unbounded codebases devolve into entropy. Establish strict architectural conventions, rigid directory structures, strongly typed contracts, and linting rules. The tighter the "chains" (constraints), the more reliably the agent performs.

- **Repo-First Context over Ephemeral Chat**

Avoid keeping critical instructions inside fleeting chat windows. Specifications, system prompts, architecture decisions, and task contracts should live version-controlled directly inside the repository (`.github/`, `docs/specs/`, `AGENTS.md`) so any agent session can resume with durable context.

- **High Code Health as an AI Prerequisite**

  AI agents struggle on deeply nested, highly coupled legacy code for the exact same reasons humans do—tangled logic burns context tokens and causes hallucinated side effects. **Refactor first to expand the AI-ready surface area** before unleashing autonomous code generators.

- **Single Source of Truth (SSOT) Nine-Layer Pyramid**

  Agents generate artifacts rapidly (code, tests, PR descriptions, specs). Without a clear hierarchy of truth, artifacts contradict each other. Enforce a single source of truth where business logic flows top-down (Mission → Strategy → Domain Model → Implementation) and updates move bottom-up backed by automated evidence.

3. Agent Execution & Verification Loops

These principles govern how agent harnesses operate in production workflows.

- **An Agent Is a Loop, Not a Model**

  The LLM is merely the reasoning engine; the *harness* (LLM + exposed tools + control flow loop + state persistence) is the actual agent. Focus engineering effort on tool design, error parsing, and state loops rather than prompt-tuning alone.

- **Start with the Dumbest Loop That Works**

  Resist the urge to prematurely build complex multi-agent graphs, orchestrator-worker trees, or dynamic state machines. Start with a single model and a few clear tools. Add complexity only when empirical evaluations prove the simple loop is failing.

- **Proof Before Claims (Verification-First Discipline)**

  Never accept an agent's confident textual claim that a task is "done". Require execution proof—green unit tests, passing type-checks, clean linters, or successful API responses—before accepting changes.

- **Context Is Your Scarcest Resource**

  Treat the LLM context window like memory on an embedded system:

  - **Curate:** Send only the specific tools and code files relevant to the active sub-task.
  - **Summarize:** Compact long step histories into structural checkpoints.
  - **Evict:** Drop heavy intermediate tool outputs once the agent reaches a conclusion.

Architectural Comparison Matrix

Principle Dimension	Legacy Software Practice	AI-Native Software Practice
Development Flow	Sequential (Spec → Code → Test → Deploy)	**Parallel by Default** (Spec, Codegen, Test, & Security run concurrently)
Primary Artifact	Source Code Syntax	**Executable Specs, Prompt Pipelines, & Eval Benchmarks**
Testing Philosophy	Unit/Integration Assertions	**Deterministic Test Gates + Probabilistic Semantic Evals**

Principle Dimension	Legacy Software Practice	AI-Native Software Practice
Refactoring Driver	Human Code Smells & Tech Debt	**Code Health Metrics to Maximize Agent Success Rates**
Error Handling	Throw Exceptions / Alert On-Call	**Close Feedback Loops (Feed Errors Back to Agent for Self-Healing)**

User prompt: are there any standard 'manifesto' / best practices generally adopted in this ai native software factory or software dev lifecycle etc.

Response: Beyond the individual 12-Factor models, the industry has begun consolidating around a few formal **manifestos** and **standard best practices** for the AI-Native Software Development Lifecycle (AI-SDLC).

When moving from individual AI autocomplete tools to full **AI software factories**, the major published manifestos and consensus practices break down into clear categories.

1. The Major Named Manifestos

A. The Agentic-Agile Manifesto (Microsoft / Open Source)

Published in open-source frameworks like Microsoft's `agentic-agile-template`, this manifesto defines how human-agent software teams collaborate.

- **Core Thesis:** *"In Software 3.0, the spec is the program."* Agents fail not because of model limitations, but because of coordination breakdowns and vague prompts.
- **The 5 Core Values:**
    1. **Specifications & Contracts** over open-ended prompts.
    2. **Human-Agent Partnership** over one-directional delegation.
    3. **Parallel Independence** over sequential handoffs.
    4. **Built-in Governance** over bolted-on compliance.
    5. **Continuous Measurement** over post-hoc assessment.

B. The AI-Native Large-Scale Agile Manifesto (arXiv)

Authored by software engineering researchers to scale agentic development across enterprise organizations.

- **Key Shift:** Replaces the meeting-heavy, document-driven waterfall/scrum process with an intelligent, adaptive workflow.
- **The 6 Core Principles:**
    - **Parallel by Default:** Planning, coding, testing, and security checks execute concurrently rather than in gated linear phases.
    - **Intent & Oversight Teams:** Engineers transition from *writing syntax* to *shaping intent, guardrails, and environment conditions*.
    - **Living Knowledge:** Replaces decaying static wikis/PDFs with version-controlled, agent-readable specs that update automatically as the codebase evolves.

- **Verification-First Assurance:** Automated quality checks replace manual approval meetings as primary release gates.
- **Orchestrated Agent Workforces:** Specialized sub-agents handle discrete tasks rather than relying on one giant monolith agent.
- **Reusable Blueprints:** Architectural patterns are codified so agents don't invent new, non-standard patterns for every task.

C. The AI-Assisted Agile Manifesto (Enterprise / Publicis Sapient)

An explicit modernization of the original 2001 Agile Manifesto:

2001 Agile Manifesto	AI-Assisted Agile Manifesto
Individuals & interactions over processes and tools	**Individuals & AI interactions** over rigid roles and ceremonies
Working software over comprehensive documentation	**Explainable, working software** over static documentation
Customer collaboration over contract negotiation	**Value validation loops** over rigid feature contracts
Responding to change over following a plan	**Continuous, automated adaptation** over fixed sprint planning

2. Generally Adopted Industry Best Practices (AI-SDLC)

Across teams running high-throughput AI software factories (such as Pulumi, GitHub Copilot Workspace, and Cursor-heavy teams), a core set of standard practices has emerged:

1. Standardized Repository Interfaces (`AGENTS.md` & MCP)

- **`AGENTS.md` File:** Just as `README.md` onboards human developers, an `AGENTS.md` file at the root of a repo acts as the entrypoint for AI coding agents. It dictates file boundaries, test commands, architectural rules, and code style without cluttering every individual prompt.
- **Model Context Protocol (MCP):** Using open standards (like Anthropic's MCP) to give agents standardized, type-safe access to issue trackers, database schemas, and CI/CD pipelines.

2. "Waves, Not Waterfalls" Execution

- Work is broken down into **zero-collision waves**. A spec decomposes a feature into small, isolated stories with explicit file-ownership boundaries so multiple agents (or human-agent pairs) can implement separate parts of a codebase simultaneously without git merge conflicts.

3. Agent-Driveable Tooling (CLI & Type-Safe APIs)

- Software factories replace web-based "click-ops" UIs with CLI-driven, type-safe infrastructure and tooling. Agents excel at running CLIs (e.g., `pulumi up`, `gh pr create`, `npm test`) and parsing structured JSON errors. If a tool lacks a CLI or API, it creates a bottleneck in the software factory.

4. Disposable Sandboxes ("Twin" Environments)

- Rather than trying to build complex multi-tenant guardrails inside shared staging environments, AI software factories spin up ephemeral, isolated cloud sandboxes (or digital twin staging environments) per agent run. Agents test and break code in disposable environments; only verified diffs hit production.

5. Shift from Code Coverage to Evaluation Metrics

- Instead of just checking line coverage percentage, AI factories measure:
  - **First-pass acceptance rate:** How often agent PRs pass test gates without human rework.
  - **Rework / Defect escape rate:** How often merged agent code introduces subtle regressions.
  - **Human hours per unit of value:** Tracking time saved rather than lines of code generated.

User prompt: i want an extremely exhaustive map of this land

Response: To map out the entire **AI-Native Software Engineering Landscape** 🗺️, we can organize all these manifestos, practices, and architectures into four primary territories that redefine the software development lifecycle.

Territory 🗺️	**Core Focus** 🎯	**Primary Standards & Manifestos** 📜
1. Agentic Architecture 🤖	How to build reliable, stateless, context-aware LLM loops	• 12-Factor Agents • 16-Factor AI Apps • Anthropic Building Effective Agents
2. The Software Factory 🏗️	How autonomous agents write, test, and ship code	• Spec-Driven Development (SDD) • Seven Rules for AI Factories • Model Context Protocol (MCP)
3. Verification & Evals 🔍	How non-deterministic output is tested and secured	• The Eval Manifesto • Self-Protecting ADLC • Continuous Benchmark CI/CD
4. Organizational Model 👥	How human teams steer, govern, and collaborate with agents	• Agentic-Agile Manifesto • AI-Native Large-Scale Agile • Intent & Oversight Roles

Together, these four domains move development from manual code writing to **Intent-Driven Orchestration**, where specifications serve as the single source of truth and agents handle execution within strict evaluation gates.

Where should we venture on this map next?

1. **Agentic Architecture Deep Dive** 🤖 : Explore the technical design rules for building reliable agent loops, tool interfaces, and context windows.
2. **Software Factory Blueprint** 🏗️ : Examine the execution mechanics, from `AGENTS.md` and MCP to disposable sandbox pipelines.
3. **Verification & Eval Frameworks** 🔍 : Look into how teams build automated evaluation gates and security controls for probabilistic code.

User prompt: AI-native software development entire lifecycle manifesto and best practices standards

Response: Here's my plan to tackle that topic. Let me know if you need to make changes.

User prompt: Start research

Response: I've completed your research. Feel free to ask me follow-up questions or request changes.

User prompt: how to transpose this AI native development standards into skills - case in point compound engineering skills

Response: Transposing AI-native software development standards into concrete human skills requires redefining what it means to be a "productive" software engineer. Traditional engineering productivity is measured in syntax output (lines of code written, speed of manual debugging, and keyboard execution speed). AI-native engineering productivity, by contrast, is measured in **system leverage**—the ability to configure, steer, and govern autonomous software-producing engines.

The definitive archetype of this transformation is **Compound Engineering**. Grounded in the work of AI-native builders like Dan Shipper, Jason Shuman, and Kieran Klaassen (Every.to), Compound Engineering is the practice of systematically capturing what works and what fails in prompts, workflows, and agent interactions so that the software factory itself gets measurably smarter over time.

Under this paradigm, instead of each added feature making a codebase more complex and harder to maintain, **each completed unit of engineering work acts as an investment that makes subsequent tasks faster and easier**. Because underlying language models do not learn between sessions, the engineer's highest-leverage skill is building and managing the persistent external memory layer that grounds the AI.

1. Syntax Writer → System Orchestrator: The Paradigm Shift

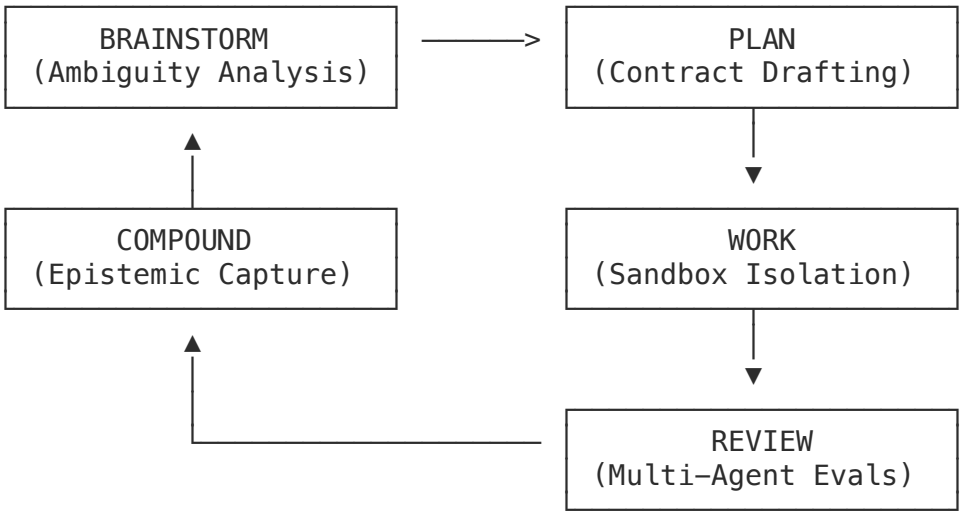
To transition to an AI-native operating model, engineers must translate legacy skills into core AI-native and compound engineering competencies:

Traditional Software Engineering Skill	AI-Native / Compound Engineering Skill	Target Toolchain & Patterns
Logic & Boilerplate Writing	**Spec-First Behavior Design**	Cursor Rules, `CLAUDE.md`, GitHub Spec Kit (`/speckit.specify`)

Traditional Software Engineering Skill	AI-Native / Compound Engineering Skill	Target Toolchain & Patterns
Manual Debugging & Code Tracing	**Isolated Sandbox Analysis & Tooling**	Ephemeral microVMs (Northflank), CLI Log Parsers
Direct Code Review (Line-by-Line)	**Multi-Agent Evaluation & Gating**	Parallel Reviewer Agents (Claude, Codex, Thursday, Friday)
Maintaining Wikis, Docs, & Readmes	**Epistemic Codification & Memory Management**	Centralized `learnings.md`, `/ce-compound`, MCP Servers
Visual "Click-ops" Configuration	**API & CLI-Driven Automation**	Declarative IaC (Pulumi TypeScript), MCP database validators

2. Deep Dive: Core Compound Engineering Skill Verticals

The Compound Engineering workflow operates as a continuous loop: **Brainstorm** → **Plan** → **Work** → **Review** → **Compound**. Master-level skills are mapped directly across these phases:



Skill Vertical A: Ambiguity Elimination & Spec-Driven Architecture (The "Brainstorm" & "Plan" Phases)

In an AI-native system, the AI acts as the muscle, but the specification is the brain. The primary skill here is **Design by Contract**—converting fuzzy human intentions into highly structured, machine-readable specifications *before* a single line of code is generated.

- **Ambiguity Analysis:** Running structured, interactive Q&A sessions with reasoning agents (using tools like `/ce-brainstorm` or Claude Code's plan mode) to intentionally fish out unstated edge cases, API dependencies, and structural boundaries.
- **Behavioral Spec Writing:** Translating business requirements into strict, behavior-oriented formats (such as Gherkin scenarios or JSON-schema specifications) that explicitly document both what is included and what is excluded from the task scope.
- **Decoupled Technical Planning:** Authorship of architectural blueprints (such as `docs/plans/`) that map the proposed data structures, C4 diagrams, and files-to-change. The engineer shifts focus from *how* to type the code to *governing* the technical plan before execution.
- *Core Philosophy:* **"80% of the effort goes into planning and review, 20% into execution."**

Skill Vertical B: Context Engineering & Sandbox Isolation (The "Work" Phase)

Because agents lack tacit human judgment and operate at high speeds, engineers must develop skills in bounding, isolating, and serving context to the AI workspace.

- **Context Window Management:** Preventing "context decay" by systematically curating and pruning token feeds. The engineer must package highly scoped "Context Packs" containing only the exact styles, domain entities, and past codebase learnings needed for the immediate sub-task.
- **State and Environment Isolation:** Running agents within secure, ephemeral containers (microVM sandboxes) or isolated git worktrees (`.worktrees/<task-id>/`). The engineer must configure the workspace so that parallel agents share only gated coordination states, never mutable working codebases.
- **Model Context Protocol (MCP) Design:** Designing and wiring custom MCP schemas to securely bridge AI agents to enterprise databases, issue trackers (Linear/Jira), and automated browser tests without exposing dangerous write keys or unmitigated access.

Skill Vertical C: Automated System Verification (The "Review" Phase)

The traditional pull request review process cannot keep pace with high-throughput agent execution. AI-native engineers focus on building automated verification networks.

- **Multi-Agent Personas Setup:** Orchestrating parallel reviewer agents to analyze code from diverse architectural viewpoints. An engineer must know how to trigger independent reviewer harnesses (e.g., configuring one agent to review exclusively for security risks, another for simplicity and SOLID principles, and a third for database performance).
- **Outcome-First Testing Architecture:** Moving away from line-by-line syntax checking and prioritizing automated, end-to-end (E2E) assertion writing (using Playwright or XcodeBuild templates). Since AI can generate structurally sound but logically misaligned code, the skill lies in asserting *system outcomes* inside ephemeral preview environments.
- *Core Philosophy:* **"Taste belongs in systems, not in review."** Ensure correctness via strict, deterministic schemas, type definitions, and continuous integration gates rather than post-hoc human rubber-stamping.

Skill Vertical D: Epistemic Codification (The "Compound" Phase)

The defining differentiator of a Compound Engineer is **durable bookkeeping**. Since models are stateless, the engineer acts as an active knowledge collector.

- **Failure Extraction:** The ability to look at a failed compilation run, test suite drop, or runtime bug, and extract the underlying architectural "why" rather than simply instructing the agent to try a different quick fix.
- **Managing the Knowledge Flywheel:** Synthesizing these failure patterns and successful code patterns into structural markdown documentation (e.g., updating a centralized repo-level `learnings.md` or adding new reference files to `docs/solutions/`).
- **Context Injection Wiring:** Configuring the codebase's configuration rules (such as `.cursorrules`, `AGENTS.md`, or `CLAUDE.md`) so that reasoning agents automatically ingest, cite, and apply these historically captured learnings at the start of every future planning session.

3. The Skill Maturity Roadmap (Stage 0 $\rightarrow$ Stage 5)

To build a team of Compound Engineers, individuals must systematically progress through six defined execution stages:

AI-SDLC SKILLS TRAJECTORY	
STAGE 0: MANUAL	Human writes every line of code by hand. Debugs with console logs and Stack Overflow [cite: 1, 3].

STAGE 1: CHAT-BASED	Uses AI as a smarter search engine. Copy-pastes snippets to and from ChatGPT or Claude [cite: 1, 3].
STAGE 2: LINE-BY-LINE	Uses in-IDE agents (Cursor Composer, Claude Code CLI). Reviews every diff line-by-line [cite: 1, 3].
STAGE 3: PLAN-FIRST	Writes detailed specs first. Lets agents execute. Reviews at PR level using automated test suites [cite: 3].
STAGE 4: IDEA-TO-PR	Describes feature in a brief. Agent plans, builds, runs browser-based QA, and submits signed PRs [cite: 3, 7].
STAGE 5: CLOUD RUNS	Orchestrates multiple agents running in parallel across isolated digital-twin environments [cite: 3, 21].

4. Immediate Practical Blueprints for Skill Adoption

To begin practicing Compound Engineering today, master these two entry-level tactical skills:

Practice A: Enforcing the `learnings.md` Flywheel

Stop starting every coding session from scratch.

1. Create a file at the root of your repository named `learnings.md`.
2. Whenever your AI coding assistant encounters a bug, a styling mismatch, or a complex integration boundary, run a "compounding" command or manually append the resolution:

   ## Learning 042: Next.js Suspense Boundary wrapping with searchParams

   - Problem: Using useSearchParams in a component without a suspense boundary de-optimizes the page to client-side rendering and crashes on static build.
   - Fix: Always wrap components utilizing searchParams inside a boundary at the page layout layer.

3. Inject this memory directly into your agent's system rules. For instance, append this instruction to your project's `CLAUDE.md` or `.cursorrules`: Before starting any code planning or task execution:
   1. Read the `learnings.md` file in the project root.
   2. Check if the active task intersects with any documented failure patterns.
   3. Explicitly state in your /plan how you will avoid repeating those documented mistakes.

Practice B: Defining the Mathematical Transition of Code Change

Master the mathematical mindset of the Phoenix Architecture:

$$\text{Desired State} = \text{Current State} + \text{Delta}[cite: 27]$$

Instead of asking an agent to "make a change," train yourself to define the change as an explicit **Delta**. When using tools like Cursor Composer or Claude Code, feed the model instructions structured exactly like this:

Goal: Implement User Profile Avatars Current State:

- DB Schema: users table contains user_id and email

- UI: User menu shows a generic icon

Delta:

- DB Schema: Run append-only migration adding avatar_url to users table
- UI: Render circular avatar image if avatar_url is not null; fallback to user's first initial

Deterministic Validation:

- Run `npm run test:db` -> MUST verify DB migration
- Run `npm run lint` -> MUST exit with code 0

By shifting your everyday interactions from conversational chatting to specifying contracts, orchestrating isolation, and systematically codifying failure patterns, you transition from a developer writing throwaway code to a Compound Engineer building a self-improving software factory.

- **Teach the system, don't do the work yourself.**
- **Type less code. Ship more value.**
